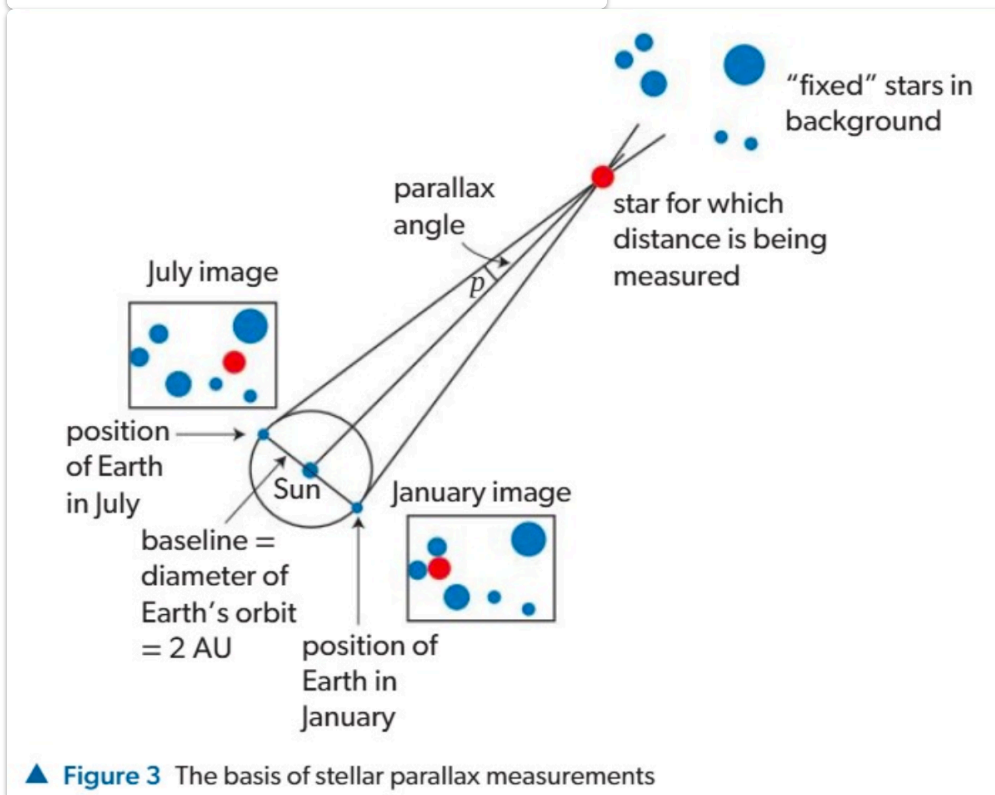
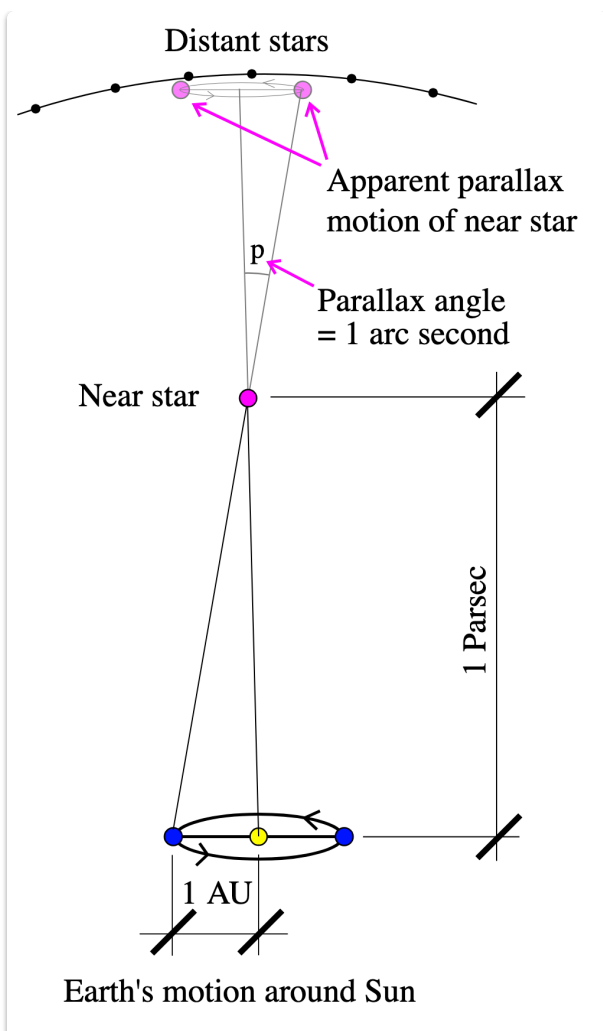


measuring astronomical distance

Parallax measurements are used to determine distances to the **nearest stars**. As the Earth moves across the diameter of its orbit over a **six-month** period, the positions of the nearest stars move relative to the 'background' of **fixed distant stars**.

Non-SI units to deal with:

- The **light year** (ly) is the distance travelled by light in one year. $1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$.
- The **astronomical unit** (AU) is the average distance between the Earth and the Sun.
 $1 \text{ AU} = 1.50 \times 10^{11} \text{ m}$.
- The **parsec** (pc) is defined using parallax angle. A star that is 1 pc from Earth will subtend^[1] a parallax angle of 1 arc-second. $1 \text{ pc} = 3.26 \text{ ly}$.



▲ **Figure 3** The basis of stellar parallax measurements

Treating Earth to be at position of July,

Distance of Earth from Sun = 1 AU

Hypotenuse = Distance from Earth to star

$$\therefore \text{Hypotenuse} = \frac{1 \text{ AU}}{\sin(p)}$$

And, due to small angle approximation, $\sin(p) \approx p$ and the data booklet of terror and despair provides the distance d to the star as:

$$d = \frac{1}{p}$$

where d is in parsec and p is in arc-seconds.

One arc-second is $\frac{1}{3600}$ of one degree (1°) and the arc-second symbol is $"$, so one milliarc-second can be written as $0.001"$.

Stellar parallax measurements from the surface of Earth allow distance estimates up to ≈ 100 pc because **turbulence in the atmosphere limits** the **smallest angle** that can be measured.

When an orbiting satellite outside the atmosphere is used, that distance goes up to 10000 ly.

-
1. In this context, subtend refers to the existence of an angle as a result of a distance. E.g a side of a triangle **subtends** the opposite angle. Subtend is latin for 'stretched under'. ↩